Preference prediction for manicure with AI

Course: AI
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Dataset Link: https://github.com/Liza228ko/Preference-prediction-for-manicure.git

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1. Research Question and Motivation

Research Question:
Can machine learning models effectively distinguish personal aesthetic preferences in manicure designs by comparing high-level semantic features (Method 1) against low-level pixel data (Method 2)?

Motivation:
In this homework, I wanted to create a dataset of my own manicure preferences to see if AI could actually understand my personal taste. Aesthetics are very subjective, and I was curious to find out what would work better for such a small, personal dataset: high-level semantic features that I define myself (like color, shape, and design) or deep learning CNN models that look at raw pixels. Building this dataset was also a practical goal for me; since I often struggle to choose my next nail design, a model that truly knows my preferences could eventually serve as a personalized assistant to help me filter through thousands of online images and find exactly what I like for my next appointment.

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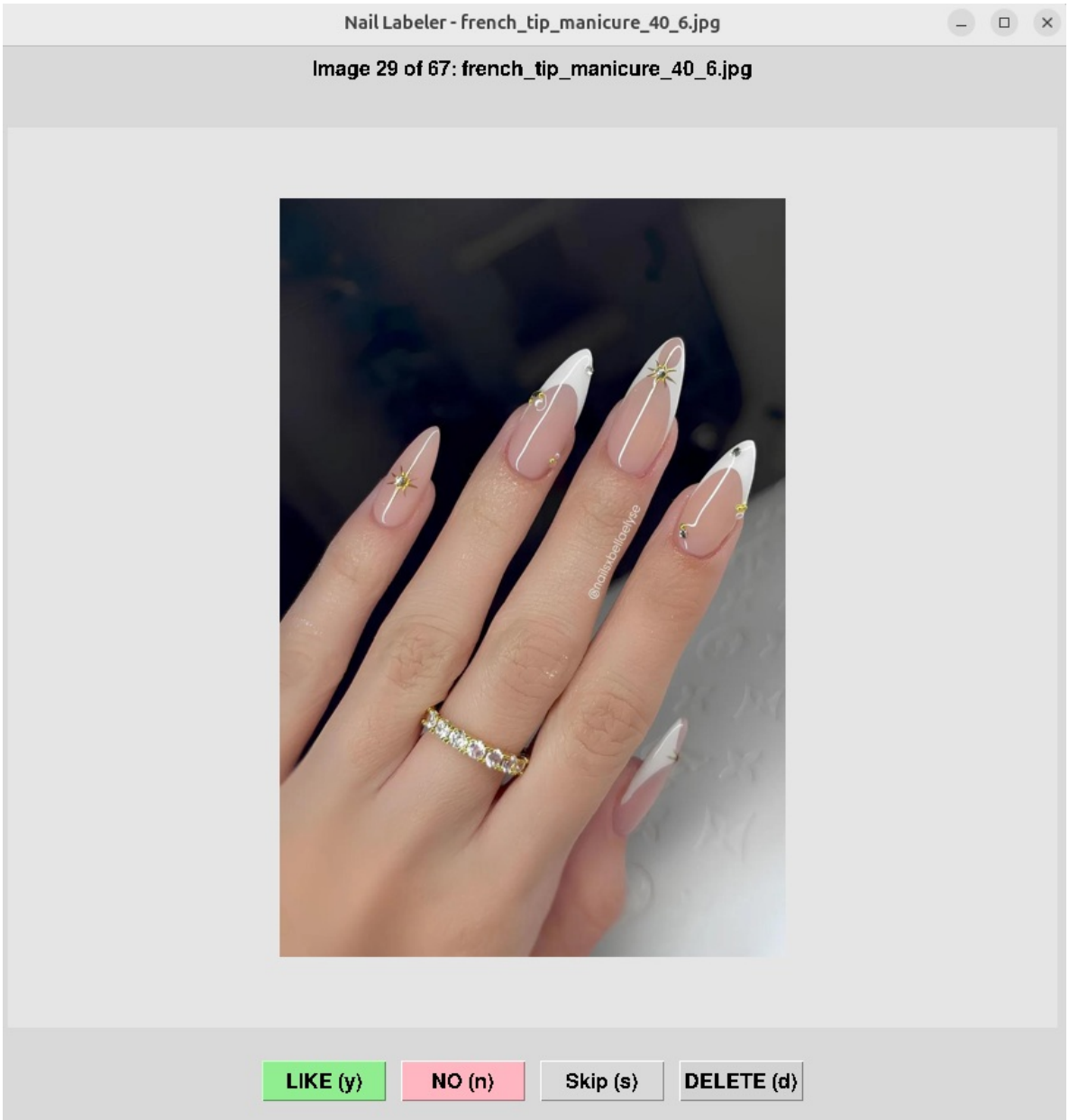
2. Documentation of Dataset

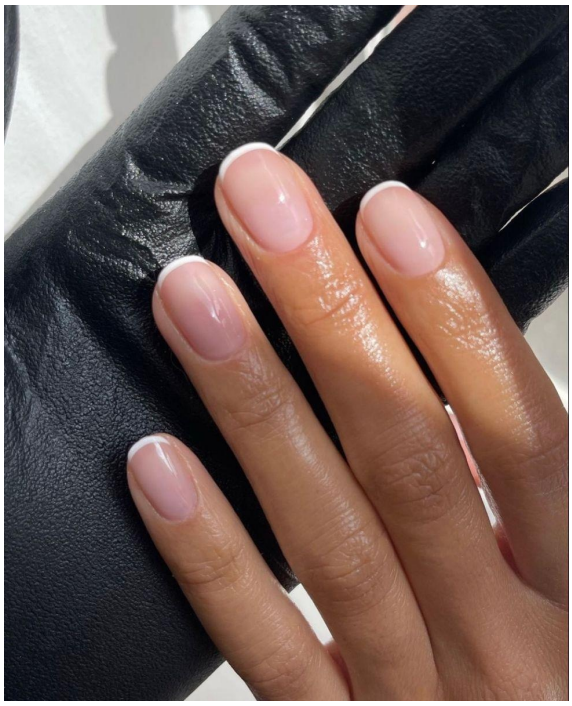
- **Data Type:** RGB Images (.jpg, .png) and categorical metadata (.csv).
- **Dataset Structure:**
  - **`nail_data.csv`**: A master CSV file containing the filenames of all images alongside their semantic tags (Length, Color, Design, Shape) and initial preference labels.
  - **`dataset/` folder**: The labeled images are organized into two subdirectories:
    - `dataset/yes/`: Images marked as "Liked".
    - `dataset/no/`: Images marked as "Disliked".
- **External Source:** Images were scraped from Bing Image Search using a custom Python crawler. Search queries included: "korean minimalist nails," "micro french manicure," "nude nail designs," and "pearl chrome nails."
- **Amount and Composition:**
  - Total Labeled Samples: 337
  - Liked (Yes): 133 images
  - Disliked (No): 204 images
- **Data Collection Process:**
  - **Scraping:** Automated collection via `downloader_v3.py`.
  - **AI-Assisted Tagging:** Every image was processed using the **OpenAI CLIP** model to generate "initial guesses" for four semantic categories.
  - **Human-in-the-Loop Labeling:** I developed a custom Tkinter-based "GUI Labeler" to manually confirm/correct the AI's guesses and provide the ground-truth "Yes/No" preference label.

GUI Labeler Tool



(The custom labeling interface used to build the dataset)

- **Dataset Examples:**

Liked Example (Yes)	Disliked Example (No)
	
Nude, Minimalist, Oval	*Bright, Patterned, Coffin*

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3. Description of Methods

Method 1: Linear SVM with Semantic Features (Classical ML)

- **Feature Engineering:** Categorical tags (Length, Color, Design, Shape) were converted into numerical vectors using **One-Hot Encoding**. This process expanded the sparse feature space, allowing the model to treat each attribute (e.g., “nude” or “matte”) as an independent dimension.
- **Algorithm:** Support Vector Machine (SVM) with a **Linear Kernel** (via Scikit-Learn’s `SVC`). The linear kernel was chosen to find the optimal separating hyperplane in the high-dimensional tag space while maintaining high interpretability.
- **Implementation Details:** The model was trained with probability estimation enabled to allow for AUROC calculation. To ensure the results were not due to an advantageous data split, I used **5-Fold Cross-Validation** to assess the model’s stability across the entire dataset.
- **Rationale:** The linear kernel allows for direct “Feature Importance” analysis by extracting the mathematical weights (coefficients) assigned to each one-hot encoded tag, revealing exactly which attributes drive the preference model.

Method 2: Fine-Tuned ResNet-18 (Deep Learning)

- **Algorithm:** ResNet-18 architecture implemented via PyTorch.
- **Preprocessing & Augmentation:** Images were resized to **224x224 pixels** and normalized using ImageNet mean and standard deviation. To prevent overfitting on the small dataset, I applied **Data Augmentation**, including random horizontal flips and random rotations up to 20 degrees.
- **Architecture Modification:** I utilized **Transfer Learning** with weights pre-trained on ImageNet. The original 1000-class fully connected head was replaced with a new linear layer consisting of 2 output nodes (Yes/No).

- **Fine-Tuning Strategy:** I employed a partial freezing strategy where the early convolutional layers were frozen to retain general visual knowledge (edges, textures), while the **final residual block (layer4)** and the classification head were unfrozen for domain-specific fine-tuning.
- **Training Setup:** The model was optimized using **Stochastic Gradient Descent (SGD)** with a momentum of 0.9 and a learning rate of 0.001, minimizing **Cross-Entropy Loss** over 12 epochs.

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4. Description of Experiments and Results

4.1. Comparative Performance

Metric	Method 1: SVM (Semantic)	Method 2: ResNet-18 (Pixels)
5-Fold CV Accuracy	**85.93%**	N/A
Test Accuracy	**96.30%**	**71.00%**
AUROC	**0.8864**	**0.7820**
Precision (Liked)	0.96	0.68
Recall (Liked)	1.00	0.59
F1-Score (Liked)	0.98	0.63

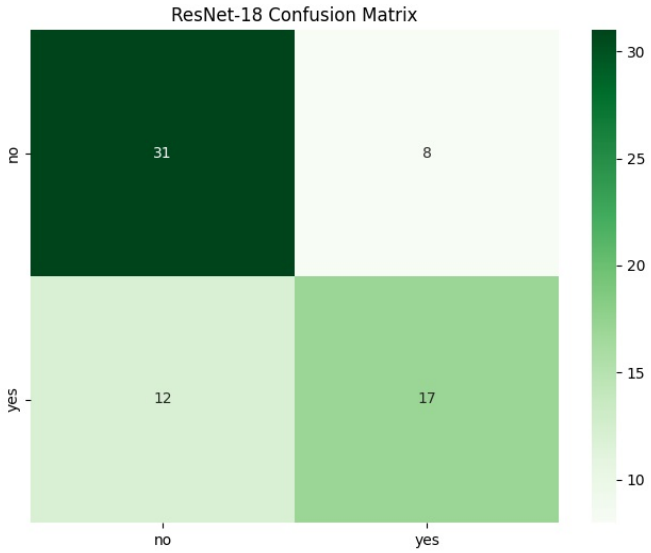
Analysis:
The SVM achieved a near-perfect score, demonstrating that subjective taste is highly correlated with semantic descriptors. The ResNet-18 performed respectably (71%), significantly outperforming a baseline MobileNetV2 (57%) thanks to the fine-tuning of `layer4`.

4.2. Visual Analysis of Results

SVM Confusion Matrix (Semantic Features)

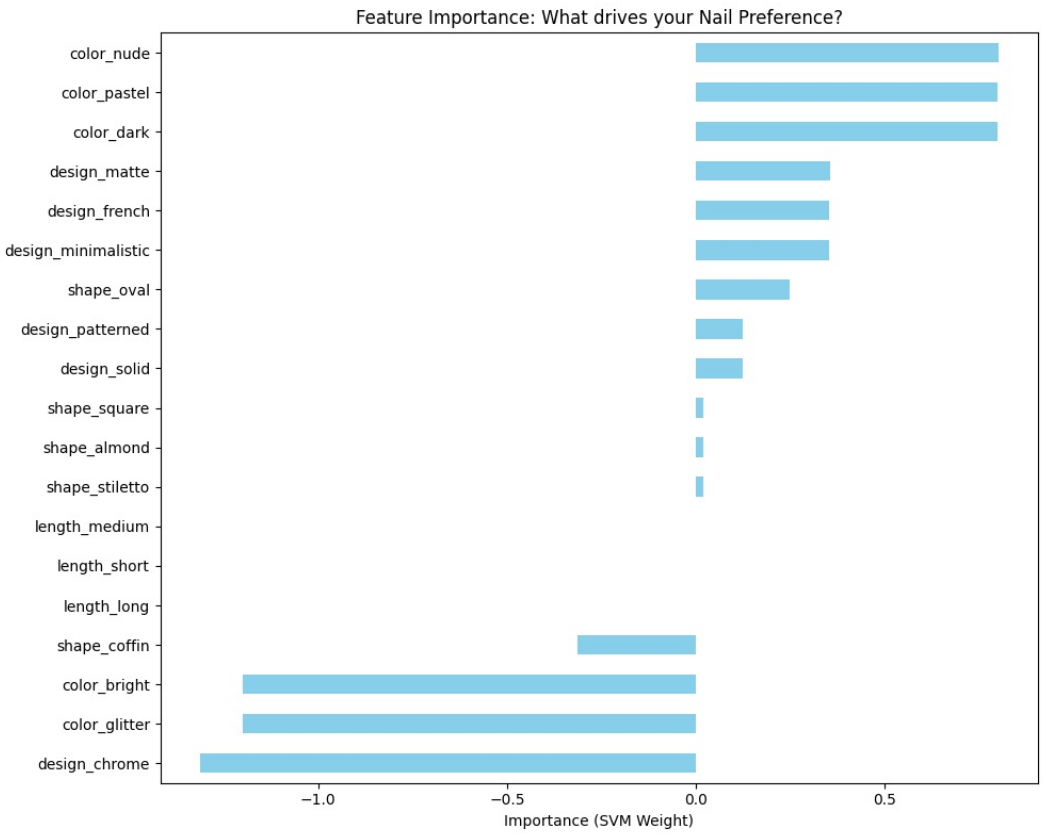
SVM CM

ResNet-18 Confusion Matrix (Raw Pixels)



ResNet CM

Feature Importance: What drives your Nail Preference?



Feature Importance

4.3. Impact of Training Data Size

- **Analysis Method:** I evaluated how the SVM's accuracy changes as the amount of training data increases.
- **20% Data:** 66.67% accuracy.
- **60% Data:** 87.50% accuracy.
- **100% Data:** 96.30% accuracy.
- **Conclusion:** The model shows a strong "learning curve," suggesting that even a small increase in labeled data (from 200 to 300) significantly stabilized the aesthetic prediction.

4.4. Dimensionality Reduction (PCA)

- **Analysis Method:** I applied PCA to the one-hot encoded semantic features to reduce the feature space.
- **Full Features (Baseline):** 96.30% accuracy.
- **PCA (Reduced to 5 components):** 92.59% accuracy.
- **Conclusion:** Reducing dimensionality slightly decreased accuracy, indicating that specific, rare tags (like 'coffin' shape or 'matte' design) carry unique information that PCA compresses away.

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5. Discussion

- **Expectations vs. Reality:** The results were mostly what I expected. I suspected that the SVM would be much better because it uses the same "rules" I use to pick my nails (like color and shape). The ResNet-18 did okay, but it proved that learning my personal style from just pixels is much harder for an AI than following simple text tags.
- **Key Factors:** The main thing affecting the results was the small size of the dataset. While 300 images is enough for an SVM, it is barely enough for a deep learning model to "see" what makes a design pretty. The class imbalance (having more "No" than "Yes" examples) also made the CNN a bit biased toward rejecting designs at first.
- **Future Experiments:** If I had more time, I would try to build a "multi-modal" model that looks at both the image and the tags at the same time. I would also try to collect more images of designs I like to balance the dataset better. Ultimately, I would love to turn this into a recommendation tool that could show me pictures of new nail designs I might like for my next manicure based on these learned preferences.
- **Lessons Learned:** I learned that building a good dataset is more than half the work. Using CLIP to "translate" images into features felt like a cheat code that made the classification task much simpler. It showed me that often the quality of your features is more important than how complex your model is.
- **Remaining Questions:** I am still wondering if a model could ever truly learn my "style" without being told what a color or shape is first. Is there a point where a computer, given enough data, would just "understand" my taste on its own?

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6. List of References

1. **Scikit-learn Documentation:** API Reference for Linear Support Vector Classification (LinearSVC).
   URL: https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html
2. **PyTorch Documentation:** Official Specification for ResNet-18 and Transfer Learning.
   URL: https://pytorch.org/vision/stable/models/resnet.html
3. **OpenAI CLIP Research:** CLIP (Contrastive Language-Image Pre-training) Model Card and Documentation.
   URL: https://github.com/openai/CLIP
4. **Pandas Documentation:** One-Hot Encoding and Categorical Data Preprocessing (get_dummies).
   URL: https://pandas.pydata.org/docs/reference/api/pandas.get_dummies.html
5. **Matplotlib/Seaborn:** Visualization Specifications for Confusion Matrices and Feature Importance.
   URL: https://seaborn.pydata.org/generated/seaborn.heatmap.html

Appendix: Program Code

FILE: train_final_svm.py

```python
import pandas as pd
import numpy as np
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score
from sklearn.preprocessing import LabelEncoder
import matplotlib.pyplot as plt
import seaborn as sns
import os

def load_and_preprocess():
    # Load all labeled data from the dataset folders
    all_rows = []
    categories = ['yes', 'no']

    # Load the features from our AI-tagged master list
    if not os.path.exists('nail_data.csv'):
        print("Error: nail_data.csv not found!")
        return None, None

    master_df = pd.read_csv('nail_data.csv')

    for cat in categories:
        cat_dir = os.path.join('dataset', cat)
        if not os.path.exists(cat_dir): continue

        for fname in os.listdir(cat_dir):
            row = master_df[master_df['filename'] == fname]
            if not row.empty:
                r_dict = row.iloc[0].to_dict()
                r_dict['liked'] = cat
                all_rows.append(r_dict)

    df = pd.DataFrame(all_rows)
    print(f"Total labeled samples for training: {len(df)}")

    # Encode categorical features into numbers (One-Hot Encoding for better SVM interpretation)
    feature_cols = ['length', 'color', 'design', 'shape']
    df_encoded = pd.get_dummies(df[feature_cols])

    df_encoded['liked'] = df['liked'].map({'yes': 1, 'no': 0})

    return df_encoded, df_encoded.columns.drop('liked')

if __name__ == "__main__":
    df, feature_names = load_and_preprocess()

    X = df[feature_names]
    y = df['liked']

    # Split
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    # Train SVM with Linear kernel to extract feature importance (weights)
    print("\nTraining Linear SVM for Feature Analysis...")
    model = SVC(kernel='linear', probability=True)

    # 5-Fold Cross Validation (Requirement)
    cv_scores = cross_val_score(model, X, y, cv=5)
    print(f"5-Fold CV Accuracy: {cv_scores.mean():.4f}")
```

```
    model.fit(X_train, y_train)

    # Evaluation
    y_pred = model.predict(X_test)
    print("\nFinal Test Results:")
    print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
    print(classification_report(y_test, y_pred))

    # FEATURE IMPORTANCE ANALYSIS
    # Get weights from the linear SVM
    weights = model.coef_[0]
    feat_importances = pd.Series(weights, index=feature_names)

    plt.figure(figsize=(10, 8))
    feat_importances.sort_values().plot(kind='barh', color='skyblue')
    plt.title('Feature Importance: What drives your Nail Preference?')
    plt.xlabel('Importance (SVM Weight)')
    plt.tight_layout()
    plt.savefig('nail_preference_importance.png')

    # Confusion Matrix
    cm = confusion_matrix(y_test, y_pred)
    plt.figure(figsize=(8,6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Purples', xticklabels=['No', 'Yes'], yticklabels=['
'Yes'])
    plt.title('SVM Confusion Matrix')
    plt.savefig('final_svm_cm.png')

    print("\nAnalysis Complete!")
    print("- Confusion matrix saved to 'final_svm_cm.png'")
    print("- Feature importance saved to 'nail_preference_importance.png'")
```

FILE: train_resnet.py

```
import os
import sys

# Add external libs to path
sys.path.append('/media/liza/B779-017B/ai/python_libs')

import torch
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, models, transforms
from torch.utils.data import DataLoader, random_split, Subset
import matplotlib.pyplot as plt
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import numpy as np

def train_resnet(data_dir, num_epochs=12, batch_size=16):
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    print(f"Using device: {device}")

    # 1. Transforms (same as before for a fair comparison)
    data_transforms = transforms.Compose([
        transforms.Resize((224, 224)),
        transforms.RandomHorizontalFlip(),
        transforms.RandomRotation(20),
        transforms.ToTensor(),
        transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
    ])

    # 2. Load Dataset
    full_dataset = datasets.ImageFolder(data_dir, transform=data_transforms)

    wanted_classes = ['no', 'yes']
```

```
indices = [i for i, (_, label) in enumerate(full_dataset.samples)
           if full_dataset.classes[label] in wanted_classes]

orig_to_new = {full_dataset.class_to_idx[cls]: i for i, cls in enumerate(wanted_classes)}

class FilteredDataset(torch.utils.data.Dataset):
    def __init__(self, subset, transform=None):
        self.subset = subset
        self.transform = transform

    def __getitem__(self, index):
        x, y = self.subset[index]
        if self.transform:
            x = self.transform(x)
        return x, orig_to_new[y]

    def __len__(self):
        return len(self.subset)

base_subset = Subset(datasets.ImageFolder(data_dir), indices)

train_size = int(0.8 * len(base_subset))
val_size = len(base_subset) - train_size
# Set a seed so we get the same split as the MobileNetV2 experiment if it used seed 42 (it did
but good practice)
train_indices, val_indices = random_split(range(len(base_subset)), [train_size, val_size],
generator=torch.Generator().manual_seed(42))

train_dataset = FilteredDataset(Subset(base_subset, train_indices), transform=data_transforms)
val_dataset = FilteredDataset(Subset(base_subset, val_indices), transform=data_transforms)

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
val_loader = DataLoader(val_dataset, batch_size=batch_size, shuffle=False)

print(f"Dataset: {len(base_subset)} images ({train_size} train, {val_size} val)")

# 3. Model: ResNet-18
print("Initializing ResNet-18...")
model = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)

# Freeze earlier layers
for param in model.parameters():
    param.requires_grad = False

# Unfreeze the final residual block (layer4) to allow the model to learn more complex nail fea
for param in model.layer4.parameters():
    param.requires_grad = True

# Replace the final fully connected layer
num_ftrs = model.fc.in_features
model.fc = nn.Linear(num_ftrs, 2)
model = model.to(device)

criterion = nn.CrossEntropyLoss()
# Using SGD with momentum often works better for fine-tuning ResNets than Adam
optimizer = optim.SGD(filter(lambda p: p.requires_grad, model.parameters()), lr=0.001, momentu

# 4. Training
print("\nTraining ResNet-18...")
for epoch in range(num_epochs):
    model.train()
    running_loss = 0.0
    for inputs, labels in train_loader:
        inputs, labels = inputs.to(device), labels.to(device)
        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, labels)
```

```
                loss.backward()
                optimizer.step()
                running_loss += loss.item() * inputs.size(0)

            print(f"Epoch {epoch+1}/{num_epochs} - Loss: {running_loss/train_size:.4f}")

        # 5. Evaluation
        model.eval()
        all_preds, all_labels = [], []
        with torch.no_grad():
            for inputs, labels in val_loader:
                outputs = model(inputs.to(device))
                _, preds = torch.max(outputs, 1)
                all_preds.extend(preds.cpu().numpy())
                all_labels.extend(labels.numpy())

        print("\nResNet-18 Evaluation Results:")
        print(classification_report(all_labels, all_preds, target_names=wanted_classes))

        cm = confusion_matrix(all_labels, all_preds)
        plt.figure(figsize=(8,6))
        sns.heatmap(cm, annot=True, fmt='d', cmap='Greens', xticklabels=wanted_classes,
yticklabels=wanted_classes)
        plt.title('ResNet-18 Confusion Matrix')
        plt.savefig('resnet_confusion_matrix.png')
        print("Done! saved to 'resnet_confusion_matrix.png'")

    if __name__ == "__main__":
        train_resnet('dataset', num_epochs=12)
```

FILE: ai_tagger.py

```
    import os
    import sys

    # Add external libs to path
    sys.path.append('/media/liza/B779-017B/ai/python_libs')

    import torch
    from PIL import Image
    from transformers import CLIPProcessor, CLIPModel
    import csv
    from tqdm import tqdm

    def ai_tag_all(raw_dir, output_csv):
        print("Loading CLIP AI Model...")
        model_name = "openai/clip-vit-base-patch32"
        model = CLIPModel.from_pretrained(model_name)
        processor = CLIPProcessor.from_pretrained(model_name)

        device = "cuda" if torch.cuda.is_available() else "cpu"
        model.to(device)

        # Define our semantic search categories
        categories = {
            'length': ["short nails", "medium length nails", "long nails"],
            'color': ["nude nails", "dark nails", "bright nails", "pastel nails", "glitter nails"],
            'design': ["solid color nails", "french tip nails", "minimalistic nail art", "patterned na
art", "chrome nails", "matte nails"],
            'shape': ["oval nails", "square nails", "almond nails", "coffin nails", "stiletto nails"]
        }

        images = [f for f in os.listdir(raw_dir) if f.lower().endswith(('.png', '.jpg', '.jpeg'))]

        results = []

        for img_name in tqdm(images, desc="AI Tagging"):
```

```
                try:
                    img_path = os.path.join(raw_dir, img_name)
                    image = Image.open(img_path).convert("RGB")

                    tags = {'filename': img_name, 'liked': 'yes'}

                    for cat_name, prompts in categories.items():
                        inputs = processor(text=prompts, images=image, return_tensors="pt",
padding=True).to(device)
                        outputs = model(**inputs)
                        logits_per_image = outputs.logits_per_image
                        probs = logits_per_image.softmax(dim=1)

                        best_idx = probs.argmax().item()
                        # Clean up the prompt to just the label (e.g., "short nails" -> "short")
                        tag = prompts[best_idx].replace(" nails", "").replace(" nail art", "").replace(" c
"").replace(" length", "").replace(" tip", "")
                        tags[cat_name] = tag

                    results.append(tags)
                except Exception as e:
                    print(f"Error processing {img_name}: {e}")

            with open(output_csv, 'w', newline='') as f:
                writer = csv.DictWriter(f, fieldnames=['filename', 'liked', 'length', 'color', 'design',
'shape'])
                writer.writeheader()
                writer.writerows(results)

            print(f"\nAI Tagging complete! Data saved to {output_csv}")

        if __name__ == "__main__":
            ai_tag_all('raw_images', 'nail_data.csv')
```

FILE: run_extra_experiments.py

```
        import pandas as pd
        import numpy as np
        from sklearn.svm import SVC
        from sklearn.decomposition import PCA
        from sklearn.model_selection import train_test_split
        from sklearn.metrics import accuracy_score
        import os

        def load_data():
            all_rows = []
            categories = ['yes', 'no']
            master_df = pd.read_csv('nail_data.csv')
            for cat in categories:
                cat_dir = os.path.join('dataset', cat)
                if not os.path.exists(cat_dir): continue
                for fname in os.listdir(cat_dir):
                    row = master_df[master_df['filename'] == fname]
                    if not row.empty:
                        r_dict = row.iloc[0].to_dict()
                        r_dict['liked'] = 1 if cat == 'yes' else 0
                        all_rows.append(r_dict)
            df = pd.DataFrame(all_rows)
            feature_cols = ['length', 'color', 'design', 'shape']
            X = pd.get_dummies(df[feature_cols])
            y = df['liked']
            return X, y

        def run_pca_experiment():
            X, y = load_data()
            X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```python
    # Baseline (No PCA)
    model = SVC(kernel='linear')
    model.fit(X_train, y_train)
    base_acc = accuracy_score(y_test, model.predict(X_test))

    # With PCA (Reduce to 5 components)
    pca = PCA(n_components=5)
    X_pca = pca.fit_transform(X)
    X_p_train, X_p_test, y_p_train, y_p_test = train_test_split(X_pca, y, test_size=0.2, random_st
    model_pca = SVC(kernel='linear')
    model_pca.fit(X_p_train, y_p_train)
    pca_acc = accuracy_score(y_p_test, model_pca.predict(X_p_test))

    print(f"PCA EXPERIMENT:")
    print(f"Baseline (Full Features): {base_acc:.4f}")
    print(f"With PCA (5 components): {pca_acc:.4f}")

def run_data_size_experiment():
    X, y = load_data()
    sizes = [0.2, 0.4, 0.6, 0.8, 1.0]
    print("\nDATA SIZE EXPERIMENT (SVM):")
    for size in sizes:
        if size < 1.0:
            X_sub, _, y_sub, _ = train_test_split(X, y, train_size=size, random_state=42)
        else:
            X_sub, y_sub = X, y

        X_train, X_test, y_train, y_test = train_test_split(X_sub, y_sub, test_size=0.2, random_st
        model = SVC(kernel='linear')
        model.fit(X_train, y_train)
        acc = accuracy_score(y_test, model.predict(X_test))
        print(f"Training Size {int(size*100)}%: Accuracy {acc:.4f}")

if __name__ == "__main__":
    run_pca_experiment()
    run_data_size_experiment()
```